

1/ $Y_1, \dots, Y_n | \mu \stackrel{iid}{\sim} N(\mu, \sigma^2)$, σ^2 is fixed and $\mu \sim N(0, \sigma^2/m)$

2/ Give a 95% posterior interval for μ .

$$\begin{aligned} A: P(\mu | Y_1, \dots, Y_n) &\propto f(Y_1, \dots, Y_n | \mu) \times \pi(\mu) \\ &\propto \prod_{i=1}^n e^{-\frac{1}{2} \frac{(Y_i - \mu)^2}{\sigma^2}} \times e^{-\frac{1}{2} \frac{m}{\sigma^2} \mu^2} \\ &\propto e^{-\frac{1}{2\sigma^2} \left[\sum_{i=1}^n (Y_i - \mu)^2 + m \mu^2 \right]} \end{aligned}$$

$$\begin{aligned} \sum_{i=1}^n (Y_i - \mu)^2 + m \mu^2 &= \sum_{i=1}^n (Y_i^2 - 2Y_i\mu + \mu^2) + m \mu^2 \\ &= (n+m)\mu^2 - 2\mu \sum Y_i + \sum Y_i^2 \end{aligned}$$

$$\text{Thus, } P(\mu | Y_1, \dots, Y_n) \propto e^{-\frac{1}{2\sigma^2} [(n+m)\mu^2 - 2\mu \sum Y_i]}$$

$$\text{thus, } \mu | Y_1, \dots, Y_n \sim N\left(\frac{\sum Y_i}{n+m}, \frac{\sigma^2}{n+m}\right)$$

Thus, a 95% posterior interval for μ can be calculated.
For example, the equal-tailed interval would be

$$\left\{ \frac{\sum Y_i}{n+m} \pm Z_{0.975} \frac{\sigma}{\sqrt{n+m}} \right\}, \text{ where } Z_{0.975} \text{ is the } 97.5\% \text{th percentile of } N(0,1) \text{ distribution.}$$

3/ select a value of m and argue that the 95% credible region is same as the 95% confidence interval.

$$A: \text{ Here, 95\% CI is } \frac{\sum Y_i}{n} \pm Z_{0.975} \frac{\sigma}{\sqrt{n}}.$$

Thus, they match when $m \rightarrow 0$.

2. Over his career, Reggie Jackson played in 2820 regular-season games and hit 563 home runs (a player can hit 0, 1, 2, ... home runs in a game). He also played in 27 World Series games and hit 10 home runs in these games. Assuming non-informative conjugate priors, summarize the posterior dist. of his home-run rate in the regular season and World Series.

$$A: Y_1, \dots, Y_{2820} \stackrel{iid}{\sim} \text{Poisson}(\lambda_1), \lambda_1 \sim \text{Gamma}(a, b)$$

$$Z_1, \dots, Z_{27} \stackrel{iid}{\sim} \text{Poisson}(\lambda_2), \lambda_2 \sim \text{Gamma}(a, b)$$

Choose a and b to be small.

So, following the derivations in the class,

$$p(\lambda_1 | Y_1, \dots, Y_{2820}) \propto f(Y_1, \dots, Y_{2820} | \lambda_1) \pi(\lambda_1)$$

$$\prod_{i=1}^{2820} e^{-\lambda_1} \frac{\lambda_1^{Y_i}}{Y_i!} \times \lambda_1^{a-1} e^{-b\lambda_1}$$

$$\propto \lambda_1^{\sum Y_i + a - 1} e^{-(2820 + b)\lambda_1}$$

$$\lambda_1 | Y_1, \dots, Y_{2820} \sim \text{Gamma}(\sum Y_i + a, b + 2820)$$

Here, $\sum Y_i = 563$.

$$\text{So, } \lambda_1 | Y_1, \dots, Y_{2820} \sim \text{Gamma}(a + 563, b + 2820)$$

$$\text{Similarly, } \lambda_2 | Z_1, \dots, Z_{27} \sim \text{Gamma}(a + 10, b + 27) \quad \left. \vphantom{\lambda_2} \right\} \text{ indep}$$

It is okay to pick $a = b = 0.1$ or 0.01 .

Q: Is there sufficient evidence to claim that he performs better in the World Series?

$$A: P(\lambda_2 > \lambda_1 | Y_1, \dots, Y_{2820}, Z_1, \dots, Z_{27})$$

$$= \int_0^\infty \int_{\lambda_1}^\infty f(\lambda_1 | Y_1, \dots, Y_{2820}) f(\lambda_2 | Z_1, \dots, Z_{27}) d\lambda_2 d\lambda_1$$

Then, need to do numerically.

3. $Y | \theta \sim \text{Exponential}(\theta)$. Find a conjugate prior for θ and derive the posterior.

A: Done in class.

$$\theta \sim \text{Gamma}(a, b)$$

$$p(\theta | Y) \propto f(Y | \theta) \pi(\theta) \propto \theta e^{-\theta Y} \times \theta^{a-1} e^{-b\theta}$$

$$\propto \theta^{1+a-1} e^{-\theta(b+Y)}$$

Thus, $\theta | Y \sim \text{Gamma}(1+a, b+Y)$.

Finding conjugate prior: Likelihood = $\theta^1 e^{-Y \cdot \theta} = \theta^{2-1} e^{-Y \cdot \theta}$

Looks like a gamma density with shape 2 and rate Y .
So, the conjugate prior is gamma.

4/ $Y|\theta \sim \text{Neg-Binomial}(\theta, m)$, and $\theta \sim \text{Beta}(a, b)$.
 $\uparrow$ Success prob. $\uparrow$ No of failures (Convention in the class)
 $Y = \# \text{ failures before } m \text{ successes}$

Q: Derive the posterior of θ .

A:
$$p(\theta|Y) \propto f(Y|\theta) \pi(\theta)$$

$$\propto \theta^Y (1-\theta)^m \times \theta^{a-1} (1-\theta)^{b-1}$$

$$\propto \theta^{Y+a-1} (1-\theta)^{m+b-1}$$

Thus, $\theta|Y \sim \text{Beta}(a+Y, b+m)$.

b) $m=5, Y=10, a=b=1$. Then plot $\theta|Y$.

A: $\theta|Y \sim \text{Beta}(1+10, 1+5) \equiv \text{Beta}(11, 6)$. For the rest, a software is needed.

In terms of R Code, it would be (equal-tailed).

$$(qbeta(0.025, 11, 6), qbeta(0.975, 11, 6)).$$

5/ In the past 50 years, an average of $\lambda_0 = 75$ large wildfires per year is observed in California. For the next 10 years, you will record the number of large wildfires and then fit a Poisson/gamma model to these data. Let the rate of large wildfires in this future period, λ , have prior $\lambda \sim \text{Gamma}(a, b)$. Select a and b so that the prior is non-informative and Variance = 100. and $\text{Prob}(\lambda > \lambda_0) = 0.5$.

A: So, $\lambda \sim \text{Gamma}(a, b)$

$$\text{Prob}(\lambda > 75) = 0.5 \Rightarrow \text{median} = 75.$$

$$\text{Var}(\lambda) = 100 \Rightarrow \frac{a}{b^2} = 100 \Rightarrow b = \frac{\sqrt{a}}{10}$$

$\lambda \sim \text{Gamma}(a, \frac{\sqrt{a}}{10})$, and $P(\lambda > 75) = 0.5$. Solve in R.

6/ An algorithm is used. It is known that the algo is unbiased, so assume that measurements follow normal dist with mean zero, $Y_i | \sigma^2 \stackrel{iid}{\sim} N(0, \sigma^2)$. If $\sigma > c$, then the algorithm must be replaced. We have $n=20$, $\sum Y_i = -2$, $\sum Y_i^2 = 15$. We conduct a Bayesian analysis with $\sigma^2 \sim \text{Inv-Gamma}(a, b)$. Compute the posterior probability that $\sigma > c$.

A: Numerically, this can be done only using a software.

$$\pi(\sigma^2) \propto (\sigma^2)^{-a-1} e^{-b/\sigma^2}$$

$$f(Y_1, \dots, Y_n | \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} Y_i^2} \propto (\sigma^2)^{-\frac{n}{2}} e^{-\frac{\frac{1}{2} \sum Y_i^2}{\sigma^2}}$$

$$\begin{aligned} p(\sigma^2 | Y_1, \dots, Y_n) &\propto f(Y_1, \dots, Y_n | \sigma^2) \pi(\sigma^2) \\ &\propto (\sigma^2)^{-\frac{n}{2} - a - 1} e^{-\frac{1}{\sigma^2} \left[b + \frac{1}{2} \sum_{i=1}^n Y_i^2 \right]} \\ &\propto (\sigma^2)^{-\left(\frac{n}{2} + a\right) - 1} e^{-\frac{1}{\sigma^2} \left[b + \frac{1}{2} \sum_{i=1}^n Y_i^2 \right]} \end{aligned}$$

$$\sigma^2 | Y_1, \dots, Y_n \sim \text{Inv-Gamma}\left(a + \frac{n}{2}, b + \frac{1}{2} \sum_{i=1}^n Y_i^2\right)$$

$$n=20, \sum Y_i^2 = 15 \Rightarrow \sigma^2 | Y_1, \dots, Y_n \sim \text{Inv-Gamma}\left(a+10, b+\frac{15}{2}\right).$$

For the rest, use R.

7/ In a simple logistic regression, the probability of a success is regressed onto a single covariate X as $p(X) = \frac{e^{\alpha + \beta X}}{1 + e^{\alpha + \beta X}}$

Assuming $X \sim N(0, 1)$, $\alpha \sim N(0, c^2)$, $\beta \sim N(0, c^2)$, find c such that the induced prior on the success prob. $p(X)$ is roughly uniform $(0, 1)$.

So, do it in R.

A:- This is a purely computational problem. You will find $c \approx 1.3$. $\alpha, \beta \sim N(0, 1.3^2)$ are not non-informative priors, as the prior variance is ~~very~~ small.

8. The Mayo Clinic conducted a study of $n = 50$ patients followed for one year ~~on~~ on a new medication and found that 30 patients experienced no ASE, 12 experienced one ASE, 6 experienced two ASE and 2 experienced ten ASE.

a) Derive the posterior of λ : $Y_1, \dots, Y_n | \lambda \stackrel{iid}{\sim} \text{Poi}(\lambda), \lambda \sim \text{Ga}(a, b)$

A: $Y_1, \dots, Y_{30} = 0, \quad Y_{43}, Y_{44}, \dots, Y_{48} = 2$

$Y_{31}, \dots, Y_{42} = 1, \quad Y_{49}, Y_{50} = 10$

$$\sum_{i=1}^{50} Y_i = 1 \times 12 + 2 \times 6 + 2 \times 10 = 12 + 12 + 20 = 44.$$

$$p(\lambda | Y_1, \dots, Y_n) \propto f(Y_1, \dots, Y_n | \lambda) \times \lambda^{a-1} e^{-b\lambda}$$

$$\propto \prod_{i=1}^n e^{-\lambda} \frac{\lambda^{Y_i}}{Y_i!} \times \lambda^{a-1} e^{-b\lambda}$$

$$\propto e^{-n\lambda} \lambda^{\sum_{i=1}^n Y_i} \cdot \lambda^{a-1} e^{-b\lambda}$$

$$\propto e^{-(n+b)\lambda} \lambda^{\sum_{i=1}^n Y_i + a - 1}$$

$$\lambda | Y_1, \dots, Y_n \sim \text{Gamma}(a + \sum_{i=1}^n Y_i, b + n)$$

(b) with $a = b = 0.01$, give the posterior mean.

A: Posterior mean = $\frac{a + \sum Y_i}{b + n} = \frac{0.01 + 44}{0.01 + 50} = \frac{44.01}{50.01}$

(c) For other a and b , the posterior mean can be similarly calculated.

(d) Plot data versus Poisson($\hat{\lambda}$) PMF. Do in R.

(e) The current medication is thought to have around one

ASE per year. Say rate = λ_2 , old medication = λ_1 .
data $Z_1, \dots, Z_n, Z_i =$

Posterior $\lambda_1 | Y_1, \dots, Y_n \sim \text{Gamma}(a + 44, b + 50)$

$\lambda_2 | Z_1, \dots, Z_n \sim \text{Gamma}(a + 50, b + 50)$

(e) What is the posterior probability that this new medication has a higher side effect?

A: ~~Pr~~ $\lambda | Y_1, \dots, Y_n \sim \text{Gamma}(a+44, b+50)$.

So, posterior probability $\Pr(\lambda > 1 | Y_1, \dots, Y_n)$

$$= \Pr(\text{Gamma}(a+44, b+50) > 1) = 1 - \text{pgamma}(1, a+44, b+50)$$

Now, we can do sensitivity analysis ~~to~~ by choosing different a and b .

9/ A Smoker is trying to quit and their goal is to abstain for one week because studies have shown that this is the most difficult period. Denote Y as the number of days before the Smoker relapses, and θ as the probability that they relapse on any given day of the study.

a) Argue that a negative binomial distribution for Y/θ is reasonable.

A: Here, ~~the~~ the event of Smoking again is a failure. And the number of days ~~until~~ until a ~~success~~ failure can be seen as a Success. Thus, a negative binomial model is reasonable.

Q: What are the most important assumptions being made and are they valid for this Smoking Cessation attempt?

A: Here, the events θ on different days are assumed to be independent. This is unlikely to be true in practice. Another, for each day, the success/failure probability remains the same. This is also unlikely.

b) Assuming a negative binomial likelihood and $\text{Unif}(0,1)$ prior for θ . Here, θ = probability that they relapse on any day. That is, θ is ~~the~~ failure probability (~~un~~ in the class, We defined negative binomial model, where θ was the success probability)

What is the prior probability ~~to~~ the Smoker will be Smoke-free for at least 7 days?

A:- $\Pr(\text{Smoker will be Smoke-free for the first } \cancel{\text{at least}} \text{ seven days})$

$$= \Pr(\text{No Smoking on day 1, } \dots, \text{ day 7})$$

$$= \Pr(\text{No Smoking on day 1}) \times \dots \times \quad \left[\begin{array}{l} \text{with our class} \\ \text{Convention} \end{array} \right]$$

$$\theta \sim U(0, 1), \quad Y|\theta \sim \text{Neg-Binomial}(1, 1-\theta)$$

$Y = \# \text{ Success before first failure.}$

$$\Pr(Y \geq 7) = \sum_{y=7}^{\infty} \Pr(Y=y)$$

$$Y \geq 7$$

$$= \sum_{y=7}^{\infty} \left(\Pr(Y=y|\theta) f(\theta) d\theta \right)$$

$\theta = \text{failure prob.}$

$$\Pr(Y=y|\theta) = \binom{1+y-1}{y} \theta^{1-y} (1-\theta)^y$$

$$= \theta^y (1-\theta)^y$$

Take, $1-\theta = \theta^*$, $\theta = 1-\theta^*$
 $\theta(1-\theta)^y = (1-\theta^*)\theta^{*y}$
 $d\theta^* = -d\theta$

$$\begin{aligned} \text{So, } \Pr(Y \geq 7) &= \sum_{y=7}^{\infty} \int_0^1 \theta^y (1-\theta)^y d\theta = \sum_{y=7}^{\infty} \int_0^1 \theta^{*y} (1-\theta^*) d\theta^* \\ &= \sum_{y=7}^{\infty} \left[\int_0^1 \theta^{*y} d\theta^* - \int_0^1 \theta^{*y+1} d\theta^* \right] = \sum_{y=7}^{\infty} \left[\left(\frac{\theta^{*y+1}}{y+1} \right) \Big|_0^1 - \left(\frac{\theta^{*y+2}}{y+2} \right) \Big|_0^1 \right] \end{aligned}$$

$$= \sum_{y=7}^{\infty} \left[\frac{1}{y+1} - \frac{1}{y+2} \right] = \sum_{y=7}^{\infty} \frac{1}{(y+1)(y+2)}$$

(c) Plot the posterior distribution of θ assuming $Y=5$.

$$Y|\theta \sim \text{Neg-Bin}(1, 1-\theta)$$

$$\pi(\theta|Y) \propto f(Y|\theta) \pi(\theta) \rightarrow \text{Unif}(0, 1)$$

$$\propto \theta^y (1-\theta)^y \times 1 = \theta^{y+1-1} (1-\theta)^{y+1-1}$$

$2, Y+1$

$$\theta|Y \sim \text{Beta}(2, 6), \text{ So, } Y=5 \Rightarrow \theta|Y \sim \text{Beta}(2, 6)$$

(d) Given this attempt resulted in $Y=5$ and that θ is the same in the second attempt, What is the posterior predictive prob. that the second attempt will result in at least 7 smoke-free days?

A:- $Y=5 \Rightarrow \theta | Y \sim \text{Beta}(2, 6)$ $Y^* = \# \text{ Smoke-free days in second attempt.}$

$$Pr(Y^* \geq 7 | Y) = \int Pr(Y^* \geq 7 | \theta) \pi(\theta | Y) d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \int Pr(Y^* \geq 7 | \theta) \times \theta^{2-1} (1-\theta)^{6-1} d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \int \sum_{y^*=7}^{\infty} Pr(Y^* = y^* | \theta) \times \theta^{2-1} (1-\theta)^{6-1} d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \sum_{y^*=7}^{\infty} \int \theta (1-\theta)^{y^*} \times \theta^{2-1} (1-\theta)^{6-1} d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \sum_{y^*=7}^{\infty} \int_0^1 \theta^{3-1} (1-\theta)^{y^*+6-1} d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \times \sum_{y^*=7}^{\infty} \frac{\Gamma(3) \Gamma(y^*+6)}{\Gamma(y^*+9)}$$

For the rest, we need a Software.